Executive Summary

The Inpatient Claims Analytics Dashboard provides a comprehensive overview of claim payment behaviors, provider performance, utilization patterns, and diagnosis-level cost drivers using a synthetic inpatient claims dataset. The analysis reveals distinct trends in how healthcare payments evolve over time and the factors influencing cost variability across providers and diagnoses.

At a glance, the dataset contains approximately 66.77K inpatient claims, representing a total claim payment volume of $639 million, with an average payment per claim of $9.58K and an average length of stay of 5.72(~6) days. The claim payment trend shows a sharp increase from early 2008, reaching its peak around late 2008, followed by a gradual decline through 2009 and 2010. By 2011, claim volumes dropped significantly from their initial highs. This pattern suggests that the early surge may have been driven by policy or reimbursement adjustments, while the subsequent decline could reflect changes in healthcare utilization patterns, reimbursement regulations, or post-recession stabilization in inpatient admissions.The breakdown of payer liability highlights that primary insurers account for about 65% of total payments, with the remaining 35% split between coinsurance and patient deductibles. This distribution emphasizes the continued financial burden patients face even in insured inpatient settings.

Provider Performance: Provider-level analysis demonstrates that a small group of hospitals or providers dominate total payment volume. For instance, providers such as 23006G, 3600NG, and 1000AH collectively receive the highest reimbursements, indicating higher patient volume or specialization in complex, high-cost procedures. When focusing on claims exceeding $20,000, the average payment per provider stands at approximately $24,570, with an average length of stay of about 9 days. Notably, providers like 03T1UA exhibit much higher averages (around $57,000) associated with longer stays of up to 24 days, suggesting case complexity as a major cost driver.

Utilization and Length of Stay Patterns: The claim utilization analysis reveals a clear skew toward shorter hospital stays, with the majority of claims corresponding to fewer than 10 inpatient days. As utilization days increase, both the count and total payment amounts decline sharply, confirming that short-term admissions dominate inpatient utilization. These patterns show that although the number of claims is high (≈66K), the majority correspond to short, low-duration stays that together accumulate to a large payment base of over $600M. However, as the length of stay increases, the average payment rises almost linearly, from around $7,000 for single-day admissions to nearly $24,000 for stays extending beyond 30 days. This demonstrates a strong correlation between duration and cost, though the relationship eventually plateaus, indicating diminishing returns for longer stays. The scatter plot between payment and utilization day count also reveals several outliers, short stays with unusually high payments, suggesting that treatment intensity or specific procedures significantly influence costs beyond just hospitalization time.

Diagnosis Level Cost Drivers: From a diagnostic perspective, the analysis identifies respiratory and cardiovascular conditions as the leading contributors to inpatient expenses. The top admitting diagnosis codes, including 78605 (Shortness of Breath), V5789 (Aftercare for Unspecified Condition), 486 (Pneumonia), 78650 (Chest Pain), and 4280 (Congestive Heart

Failure), collectively account for over $300 million in total claim payments. A treemap visualization of admitting diagnosis codes reinforces that respiratory distress and cardiac-related admissions dominate the dataset, aligning with national trends where these conditions represent major cost drivers for inpatient care. Further, the matrix visualization cross-referencing the top 10 providers against diagnosis codes shows a concentration of claim payments within a handful of diagnoses, illustrating both specialization and reimbursement concentration among top providers.

Drivers of Claim Payment Variation: The decomposition analysis adds further depth, uncovering the interaction between diagnosis, deductible amount, and length of stay as key determinants of payment. For example, claims associated with 7101(hypertensive vascular diseases) show significant payment variation when deductible levels and utilization days are factored in, with the highest observed payments exceeding $57,000 for stays of around 8 days. These findings highlight how multiple claim-level attributes interact to drive cost, offering a framework for predictive modeling and reimbursement forecasting.

In summary, the dashboard paints a holistic picture of inpatient claim behavior: payments are highly concentrated among a few providers and diagnoses, short stays dominate utilization, and cost increases are largely proportional to length of stay but strongly influenced by case complexity. The consistent dominance of cardiopulmonary diagnoses underscores the need for preventive and chronic care management strategies to reduce hospital burden. Financially, the analysis reveals that while insurers carry most of the payment responsibility, patient cost sharing through deductibles and coinsurance remains substantial.

These insights have practical implications for hospital administrators, payers, and policymakers. They can guide resource allocation, reimbursement optimization, and cost-control strategies, while also enabling the development of predictive analytics to identify cost outliers and forecast payment behavior. By integrating these findings into quality improvement and financial planning processes, healthcare organizations can better balance cost efficiency with patient outcomes.

References:
Code book: https://www.cms.gov/files/document/de-10-codebook.pdf-0
Data:
https://www.cms.gov/research-statistics-data-and-systems/downloadable-public-use-files/synpufs/downloads/de1_0_2008_to_2010_inpatient_claims_sample_1.zip
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